

Numerical Problems on Comparative and Absolute Advantage

Problem 1: Identifying Absolute Advantage

Two countries, Alpha and Beta, produce only two goods: cars and computers. The table below shows the number of units each country can produce with the same amount of resources:

Country	Cars (units)	Computers (units)
Alpha	50	100
Beta	60	80

Questions:

1. Which country has the absolute advantage in the production of cars?
2. Which country has the absolute advantage in the production of computers?

Answers:

1. Beta has the absolute advantage in cars because it can produce 60 units compared to Alpha's 50.
2. Alpha has the absolute advantage in computers because it can produce 100 units compared to Beta's 80.

Problem 2: Opportunity Costs and Comparative Advantage

The following table shows the production capabilities of two countries, Country X and Country Y, if they devote all their resources to producing one good:

Country	Wheat (tons)	Cloth (units)
Country X	40	20
Country Y	30	15

Questions:

1. What is the opportunity cost of producing 1 ton of wheat in Country X?
2. What is the opportunity cost of producing 1 ton of wheat in Country Y?
3. Which country has the comparative advantage in wheat production?

4. Which country has the comparative advantage in cloth production?

Answers:

1. In Country X, 1 ton of wheat costs $\frac{20}{40} = 0.5$ units of cloth.
2. In Country Y, 1 ton of wheat costs $\frac{15}{30} = 0.5$ units of cloth.
3. Neither country has a comparative advantage in wheat production as their opportunity costs are equal.
4. Neither country has a comparative advantage in cloth production either, as the opportunity costs are identical.

Problem 3: Specialization and Trade

Two individuals, Alice and Bob, produce bread and cheese. The table below shows how much of each good they can produce in a day:

Individual	Bread (loaves)	Cheese (blocks)
Alice	8	16
Bob	6	12

Questions:

1. Calculate the opportunity cost of producing 1 loaf of bread for both Alice and Bob.
2. Who has the comparative advantage in producing bread?
3. If Alice and Bob specialize and trade, who should produce which good?

Answers:

1.
 - Alice: 1 loaf of bread costs $\frac{16}{8} = 2$ blocks of cheese.
 - Bob: 1 loaf of bread costs $\frac{12}{6} = 2$ blocks of cheese.
2. Neither Alice nor Bob has a comparative advantage in producing bread because their opportunity costs are the same.
3. Since opportunity costs are equal, the decision on specialization would need to factor in externalities.